AI-Based Clinical Decision Support System

DBMS Projects – Architecture & Frontend Overview

Prof. ACS Rao

Associate Professor
Indian Institute of Technology (Indian School of Mines)

Project Objective

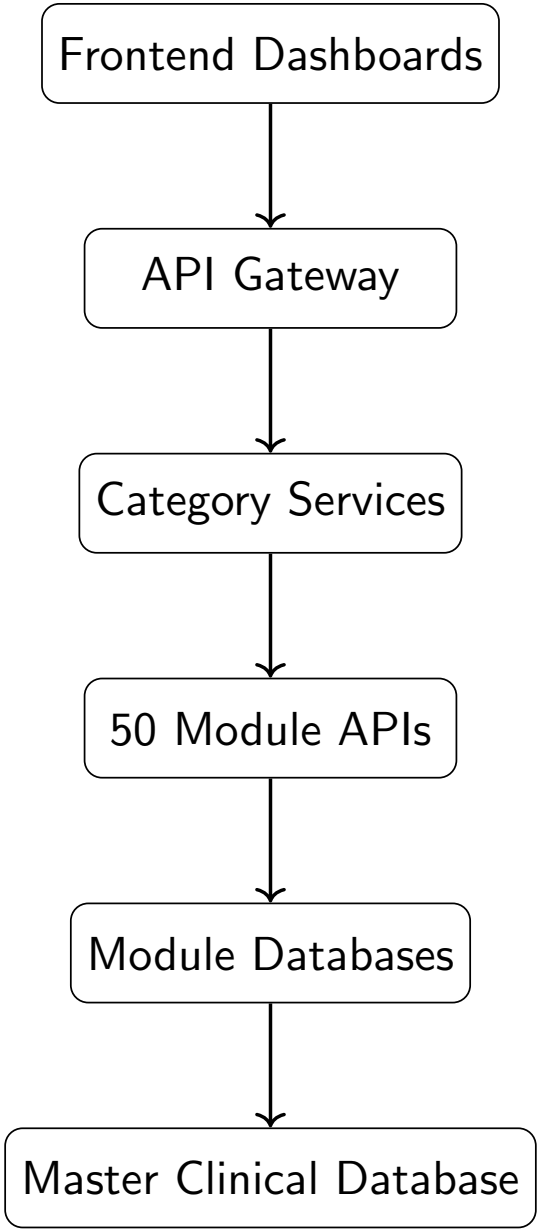
- Build a **Database-Centric Clinical Decision Support System**
- Focus strictly on **core DBMS concepts**
- No Machine Learning – logic via:
  - SQL Queries
  - Views
  - Triggers
  - Stored Procedures
- Large-scale system will be developed by **50 independent groups with 3 students in each group.**
- The details of the project are attached at last.

Overall System Philosophy

Key Design Idea

- Modular Architecture
- Each team develops one independent module
- Modules are grouped into Categories (A–I)
- Categories connect to a Master Dashboard
- All data integrates into a Master Database

System Architecture Overview

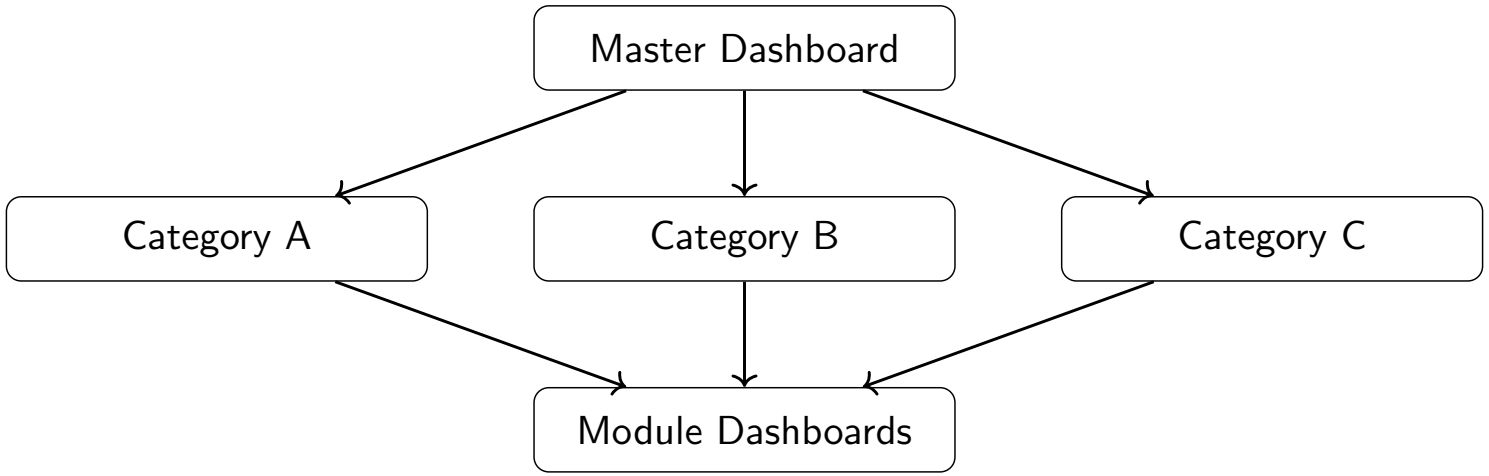


Frontend Structure

Frontend is divided into **3 hierarchical levels**

1. Master Dashboard
2. Category Dashboards
3. Module Dashboards

Frontend Hierarchy Diagram



Master Dashboard

Purpose:

- Overall system view
- Used for demonstration and evaluation

Features:

- Category-wise statistics
- Global patient index
- System-wide alerts
- Audit logs

Category Dashboards

Each Category (A–I) has:

- Overview metrics
- List of assigned modules
- Aggregated SQL views
- Rule-based alerts

Example Categories:

- Patient Data Management
- Drug Safety Systems
- ICU Monitoring
- Laboratory Interpretation

Module Dashboards (Team Responsibility)

Each Team Develops:

- One independent module
- One database schema
- One backend API
- One frontend dashboard

Module UI Must Show:

- Tables
- SQL Queries
- Triggers / Procedures
- Output results

Frontend Module Contribution Guidelines (Students)

Repository Access:

- Work using a FORK of the main repository
- No direct access to organization repository

Team Setup (3 students):

- One member forks the repository
- Others added as collaborators to the fork

Mandatory Workflow:

1. Fork organization repository
2. Clone your fork locally
3. Create branch: `module/<module-name>`

Frontend Module Contribution Guidelines (Contd.)

Folder Rule:

- Work ONLY inside: `src/modules/<module-name>/`
- Do NOT modify other modules without approval

Commits:

- ALL team members MUST make commits
- Meaningful commit messages required
- Good: "Add login form UI"
- Bad: "final", "update", "done"

Push & Pull Request:

- Push module branch to your fork
- Create PR to organization repository
- One module = One Pull Request

Frontend Module Contribution Guidelines (Contd.)

PR Requirements:

- Module name
- Team member names + GitHub IDs
- List of features implemented
- Individual contribution summary

Resources:

- **GitHub Organization:** `https://github.com/Medical-Copilot-System/Frontend.git`
- **Tutorial Video:** `https://youtu.be/q8EevlEpQ2A`

Frontend Module Evaluation Guidelines (Teachers)

Submission Format:

- All work via Pull Requests (PRs)
- Pull Requests = Assignments

Primary Evaluation Tool:

- Files changed
- Lines added/removed
- Commit history
- Individual contributors

Checkpoints for Evaluation:

- File changes restricted to assigned module
- All team members have commits
- Commit quality and messages

Frontend Module Evaluation Guidelines (Contd.)

Contribution Verification:

- Use PR $\rightarrow$ Commits to assess individual effort
- Cross-check PR description with commit history
- Filter commits by author if needed

Cheating/Free-riding Indicators:

- Only one student committing
- Identical code across teams
- Last-minute bulk commits
- Copy-paste patterns

Feedback Method:

- Use PR comments for technical feedback
- Request changes if necessary
- Approve PR only after verification

Frontend Module Evaluation Guidelines (Contd.)

Recommended Grading Rubric:

- Code correctness & completeness: 30%
- Individual contribution (commits): 25%
- Pull Request quality: 20%
- Folder & module discipline: 15%
- Commit hygiene & clarity: 10%

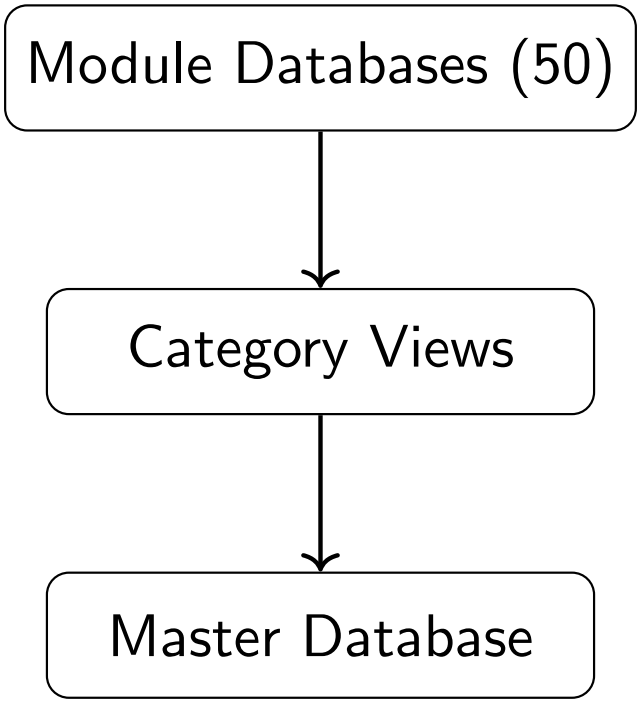
Record Keeping:

- PR links as submission proof
- GitHub history as plagiarism-safe evidence

Optional Enhancements:

- Protect main branch
- Use GitHub Insights → Contributors

Backend Database Architecture



Master Database Concept

Master DB Contains:

- Unified patient registry
- Cross-module views
- Global alerts
- Audit and access logs

Populated Using:

- SQL Views
- Triggers
- Stored Procedures

Integration Strategy

1. Each team completes module independently
2. APIs validated at category level
3. Data exposed via controlled views
4. Final integration into Master Database

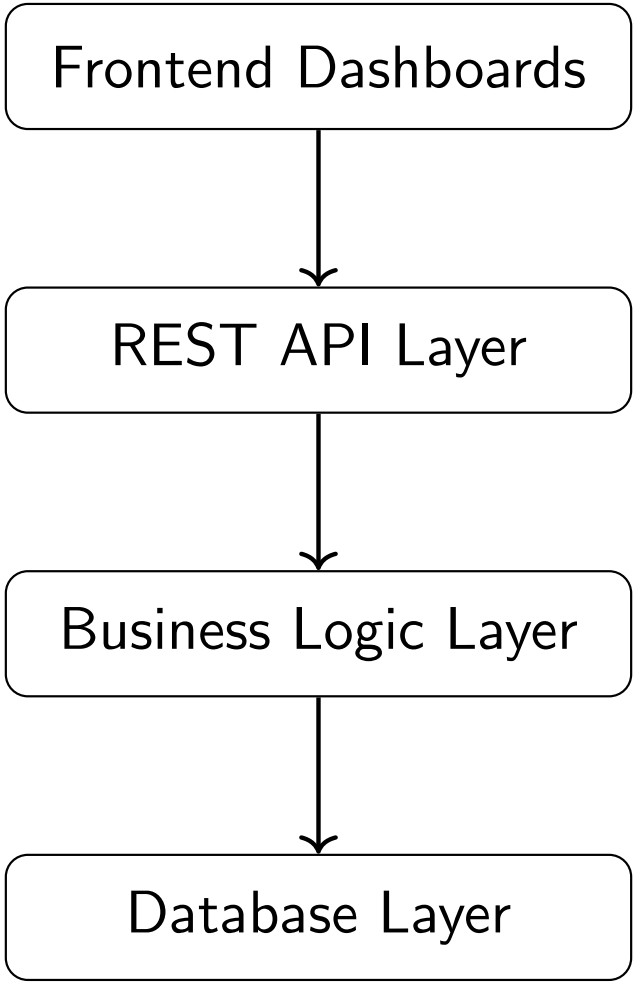
What Will Be Evaluated

- Database design and normalization
- Use of constraints
- Quality of SQL queries
- Triggers and procedures
- Integration correctness
- Understanding during viva

Backend Architecture Overview

- Backend follows a **modular service-oriented architecture**
- Each module has:
  - Its own database
  - Its own backend service (API)
- Category services aggregate multiple modules
- Master backend connects all categories

Backend Layered Architecture



Module-Level Backend Responsibility

Each Team Must Implement:

- One backend service (API)
- Database connection logic
- CRUD operations using SQL
- Triggers / Procedures in DB

Important Rule:

Isolation Principle

Modules must NOT directly access other module databases.

API Integration Strategy

Why APIs?

- To allow controlled communication between modules
- To enable category-level aggregation
- To support master dashboard integration

API Types:

- Module APIs
- Category APIs
- Master APIs

API Communication Flow

Database Integration Strategy

Integration is done using DBMS concepts only:

- SQL Views
- Stored Procedures
- Triggers

Flow:

1. Module DB generates validated data
2. Exposed through views or procedures
3. Pulled into Master DB

Role of Master Database

- Acts as the single source of truth
- Maintains global patient and visit indexes
- Enables cross-category queries
- Supports master dashboard analytics

What You Must Learn – Backend Concepts

- Client–Server Architecture
- RESTful APIs (GET, POST, PUT, DELETE)
- Database Connectivity (JDBC / Python / PHP)
- API Testing (Postman / curl)

What You Must Learn – DBMS Concepts

- ER Modeling and Normalization
- Constraints and Referential Integrity
- SQL Queries (Basic to Advanced)
- Views and Materialized Views
- Triggers and Stored Procedures

What You Must Learn – Integration Skills

- How APIs fetch data from databases
- How multiple services integrate safely
- Data validation and consistency
- Error handling in backend systems

Common Mistakes to Avoid

- Writing SQL without normalization
- Hardcoding data instead of querying DB
- No use of triggers or procedures
- Direct DB access between modules
- Poor naming conventions

Final Instructions to Students

Important

- Think beyond your single module
- Design for integration
- Follow DBMS best practices
- Be ready to explain every design choice

Final Takeaway

Think Like System Designers

- Your module is part of a larger system
- Design clean schemas
- Follow naming conventions
- Ensure integration readiness

Evaluation Summary

Pull Requests + Commits together provide a transparent, fair, and scalable evaluation system.

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROJECT ALLOTMENTS

Academic Year: 2025–2026
Semester: Winter
Course Title: Database Management Systems (DBMS)
Course Code: _____
Course Coordinator: Prof. ACS Rao

Subject

Allotment of DBMS Projects on AI-Based Clinical Decision Support / Medical Copilot Systems

This is to inform all students that, as part of the **Database Management Systems (DBMS)** course curriculum, a **group-based mini project** has been allotted for the current semester.
The objective of these projects is to provide students with practical exposure to **real-world database design and implementation**, aligned with **Outcome-Based Education (OBE)**. All projects are centered on **AI-inspired Clinical Decision Support Systems**, with emphasis strictly on **core DBMS concepts**.

Project Theme

AI-Based Clinical Decision Support Systems
(Note: AI logic is to be implemented using SQL queries, views, triggers, procedures, and rule-based logic only. No machine learning implementation is required.)

Group Formation Guidelines

- Each group shall consist of a maximum of **3 students**
- Each group will be allotted **one unique project module**
- Duplication of problem statements across groups is **not permitted**

Project Modules

Category A: Patient Clinical Data Management

1. Patient Demographics & Visit History Database
2. Chronic Disease Patient Record Management
3. Pediatric Patient Clinical Data System
4. Geriatric Patient Health Record Database
5. Patient Allergy & Immunization Tracking System
6. Clinical Alert System for Abnormal Vital Values

Category B: Symptom–Disease Diagnosis Support

7. Symptom–Disease Mapping Database
8. Fever-Based Differential Diagnosis System
9. Respiratory Symptom Diagnosis Database
10. Gastrointestinal Disorder Diagnosis Support
11. Neurological Symptom Analysis Database
12. Rule-Based Disease Ranking System

Category C: Clinical Query Copilot (Natural Language to SQL)

13. Natural Language Patient Search System
14. Clinical Query Translator for Laboratory Records
15. Voice-Assisted Clinical Query System (Text Simulation)
16. Doctor-Friendly SQL Query Dashboard
17. Smart Clinical Views using SQL
18. Question–Answering System for Hospital Database

Category D: Drug & Prescription Safety Systems

19. Drug–Drug Interaction Alert Database
20. Prescription Validation & Consistency System
21. Allergy-Aware Medication Alert System
22. Polypharmacy Risk Detection Database
23. High-Risk Drug Monitoring System
24. Automated Prescription Audit System

Category E: ICU & Real-Time Monitoring Databases

25. ICU Vital Signs Monitoring Database
26. Emergency Room Patient Alert System
27. Cardiac ICU Monitoring Database
28. Neonatal ICU Monitoring System
29. Threshold-Based Clinical Alert Database
30. Time-Series Patient Health Data System

Category F: Case-Based Clinical Decision Support

31. Historical Case Comparison Database
32. Treatment Outcome Analysis System
33. Disease Progression Case Repository
34. Readmission Risk Analysis Database
35. Therapy Effectiveness Evaluation System
36. Similar Patient Case Retrieval System

Category G: Secure EHR & Access Control Systems

37. Secure Electronic Health Record (EHR) Database
38. Role-Based Access Control for Hospital Systems
39. Clinical Audit Trail & Logging System
40. Patient Consent & Data Privacy Database

41. Multi-Role Access Control System
42. Secure Clinical Summary View Generator

Category H: Laboratory Test Interpretation Systems

43. Laboratory Test Management Database
44. Automated Lab Result Interpretation System
45. Reference Range Validation Database
46. Follow-Up Test Recommendation System
47. Pathology Report Management Database
48. Critical Lab Value Alert System

Category I: Integrated Capstone-Style Mini Projects

49. Integrated Clinical Decision Support Database (Patients, Symptoms, Drugs, Labs)
50. AI-Inspired Medical Copilot using DBMS Concepts

Submission Requirements

Each group must submit:

- ER Diagram
- Normalized Database Schema
- SQL Scripts (DDL, DML, Constraints)
- Sample Queries and Outputs
- Triggers / Procedures / Views
- Project Report
- Demonstration and Viva (if possible)

Important Instructions

- Only clinical data must be used.
- Plagiarism or duplication will lead to disqualification.
- Project must strictly adhere to DBMS concepts.

Course Coordinator: **Signature:**

Date: _____ **Place:** _____

PROJECT ARCHITECTURE FLOW DIAGRAM

AI-Based Clinical Decision Support Systems

Module Relationships and Dependencies

Academic Year: 2025-2026 — Winter Semester

Detailed Module Dependency Matrix

Data Flow Architecture

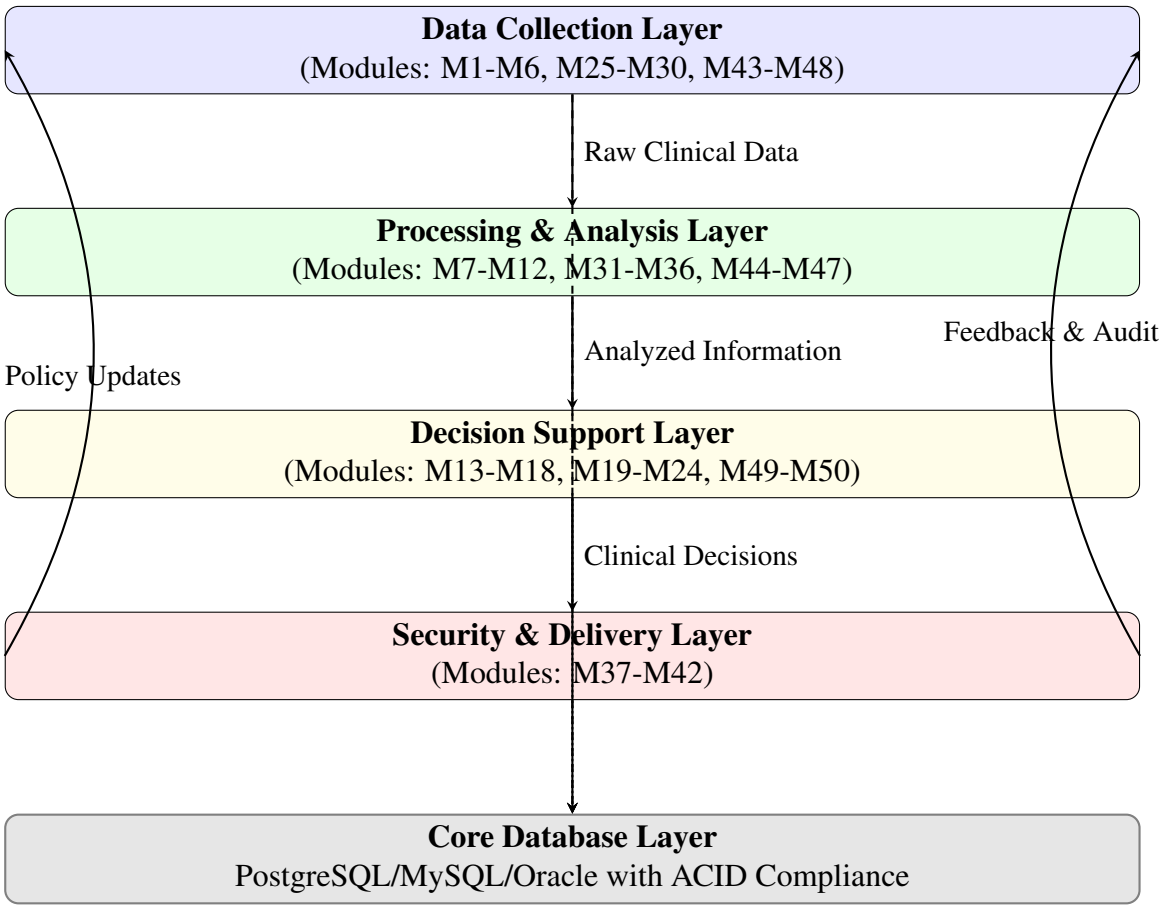


Figure 1: Four-Tier Data Flow Architecture

System Integration Points

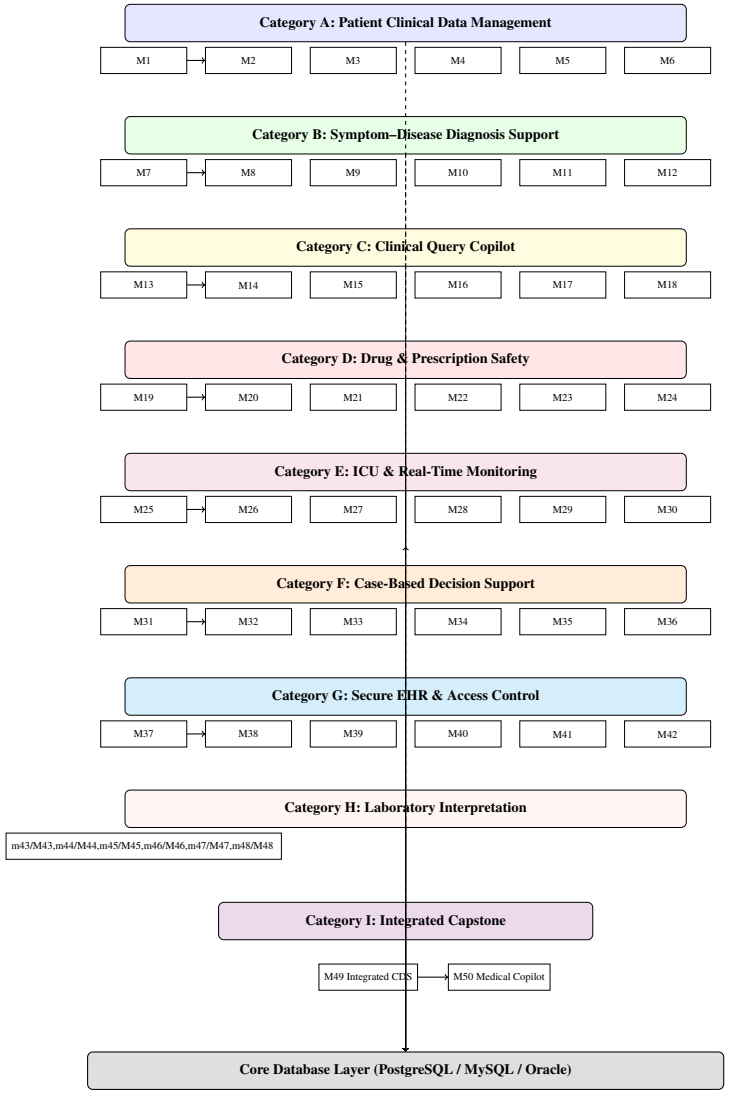
Comprehensive System Architecture Flow Diagram

Module	Module Name	Dependent On	Provides Input To
M1	Patient Demographics	Core DB	M2-M6, M7, M13, M19, M25, M31
M2	Chronic Disease Management	M1	M19, M25, M33
M3	Pediatric System	M1	M19, M28
M4	Geriatric System	M1	M19, M22, M34
M5	Allergy Tracking	M1	M19, M21
M6	Clinical Alerts	M1	M25, M29, M48
M7	Symptom-Disease Mapping	M1	M8-M12, M13, M19
M8	Fever Diagnosis	M7	M13, M25
M9	Respiratory Diagnosis	M7	M13, M27
M10	GI Disorder Diagnosis	M7	M13, M44
M11	Neurological Symptom	M7	M13, M27
M12	Rule-Based Ranking	M7	M13, M44, M50
M13	Natural Language Search	M1, M7	M14-M18
M14	Lab Query Translator	M13	M15, M44
M15	Voice Assistant	M14	M16, M50
M16	Doctor Dashboard	M15	M17, M49
M17	Smart Clinical Views	M16	M18, M42
M18	QA System	M17	M49
M19	Drug Interaction	M1, M7	M20-M24
M20	Prescription Validation	M19	M21, M24
M21	Allergy-Aware Medication	M5, M20	M22, M23
M22	Polypharmacy Risk	M4, M21	M23, M34
M23	High-Risk Drug Monitor	M21, M22	M24, M25
M24	Prescription Audit	M20, M23	M37
M25	ICU Vital Signs	M1, M6, M23	M26-M30
M26	ER Patient Alert	M25	M27, M29
M27	Cardiac ICU	M9, M11, M26	M28, M30
M28	Neonatal ICU	M3, M27	M29, M30
M29	Threshold-Based Alerts	M6, M26, M28	M30, M48
M30	Time-Series Analysis	M27, M28, M29	M36, M44
M31	Historical Case Compare	M1, M25	M32-M36
M32	Treatment Outcome	M31	M33, M35
M33	Disease Progression	M2, M32	M34, M35
M34	Readmission Risk	M4, M22, M33	M35, M49
M35	Therapy Effectiveness	M32, M33, M34	M36, M49
M36	Similar Patient Retrieval	M30, M35	M49, M50
M37	Secure EHR Database	M24	M38-M42
M38	Role-Based Access	M37	M39, M41
M39	Clinical Audit Trail	M38	M40, M42
M40	Patient Consent Privacy	M39	M41, M42
M41	Multi-Role Access	M38, M40	M42
M42	Secure Clinical Views	M17, M39, M41	M49
M43	Lab Test Management	Core DB	M44-M48
M44	Automated Lab Interpretation	M10, M12, M14, M30, M43	M45, M46
M45	Reference Range Validation	M44	M46, M47
M46	Follow-Up Test Recommendation	M44, M45	M47, M48
M47	Pathology Report Management	M45, M46	M48
M48	Critical Lab Value Alert	M6, M29, M46, M47	M49
M49	Integrated CDS Database	M16, M18, M34-M36, M42, M48	M50
M50	AI Medical Copilot	M12, M15, M36, M49	Core DB

Table 1: Module Dependency Relationships

Integration Point	**Modules Involved**	**Data Exchange**
Patient Data Hub	M1, M7, M13, M19, M25, M31	Patient demographics, visit history, basic medical information
Diagnostic Engine	M7-M12, M44-M47	Symptoms, lab results, diagnostic rules, interpretation logic
Medication Safety	M19-M24, M5, M22	Drug interactions, allergies, prescriptions, monitoring data
Critical Care	M25-M30, M6, M48	Vital signs, alerts, monitoring parameters, time-series data
Clinical Intelligence	M31-M36, M13-M18	Case histories, treatment outcomes, similarity algorithms
Security Framework	M37-M42, All modules	Access controls, audit trails, consent management, data privacy
Query Interface	M13-M18, M49-M50	Natural language queries, search patterns, results formatting

Table 2: Key System Integration Points


Category A: Patient Clinical Data Management
M1
M2
M3
M4
M5
M6
Category B: Symptom–Disease Diagnosis Support
M7
M8
M9
M10
M11
M12
Category C: Clinical Query Copilot
M13
M14
M15
M16
M17
M18
Category D: Drug & Prescription Safety
M19
M20
M21
M22
M23
M24
Category E: ICU & Real-Time Monitoring
M25
M26
M27
M28
M29
M30
Category F: Case-Based Decision Support
M31
M32
M33
M34
M35
M36
Category G: Secure EHR & Access Control
M37
M38
M39
M40
M41
M42
Category H: Laboratory Interpretation
m43/M43,m44/M44,m45/M45,m46/M46,m47/M47,m48/M48
Category I: Integrated Capstone
M49 Integrated CDS
M50 Medical Copilot
Core Database Layer (PostgreSQL / MySQL / Oracle)

Implementation Sequence Recommendations

Phase 1: Foundation Modules (Weeks 1-4)

- **M1: Patient Demographics** - Core foundation for all systems
- **M43: Lab Test Management** - Basic laboratory data structure
- **M37: Secure EHR Database** - Security framework setup

Phase 2: Specialized Modules (Weeks 5-8)

- Choose one specialty area:
  - **Clinical:** M2-M6, M25-M30
  - **Diagnostic:** M7-M12, M44-M48
  - **Medication:** M19-M24
  - **Decision Support:** M31-M36
- Implement corresponding query modules: M13-M18

Phase 3: Integration Modules (Weeks 9-12)

- **M49: Integrated CDS Database** - Combine selected modules
- **M42: Secure Clinical Views** - Unified interface
- **M50: AI Medical Copilot** - Advanced integration

Phase 4: Testing & Optimization (Weeks 13-14)

- Cross-module integration testing
- Performance optimization
- Security auditing
- User acceptance testing

Key Architectural Principles

- **Modularity:** Each module functions independently yet integrates seamlessly
- **Scalability:** Architecture supports adding new modules without redesign

- **Security by Design:** Security modules (M37-M42) integrate with all other modules
- **Data Consistency:** Core database ensures ACID compliance across all modules
- **Interoperability:** Standardized data formats and interfaces between modules
- **Extensibility:** Easy to add new clinical domains or functionality

COMPLETE PROJECT MODULE SPECIFICATIONS

Modules 1-50: AI-Based Clinical Decision Support Systems

Database Management Systems Implementation

Academic Year: 2025-2026 — Winter Semester

Module 1: Patient Demographics & Visit History Database

Primary Objectives:

- Design comprehensive patient registration system with demographic validation
- Implement temporal tracking of patient visits with audit trails
- Create intelligent appointment scheduling with conflict detection
- Develop referral network management with status tracking

Technical Specifications:

- **Entities:** Patient, Visit, Appointment, Department, Physician, Referral
- **Key Attributes:** Patient (UUID, demographics, insurance), Visit (visit_id, date, reason, diagnosis, status)
- **Relationships:** One-to-Many (Patient-Visit), Many-to-Many (Visit-Department)
- **Advanced Features:** Automatic age calculation trigger, visit frequency analysis, no-show prediction
- **SQL Requirements:** 25+ queries including temporal joins, cohort analysis, pattern recognition

Module 2: Chronic Disease Patient Record Management

Primary Objectives:

- Design specialized database for chronic disease progression tracking
- Implement medication adherence monitoring with reminder systems
- Create disease-specific assessment score calculations
- Develop complication prediction algorithms using historical data

Technical Specifications:

- **Entities:** ChronicDisease, DiseaseEpisode, TreatmentPlan, MedicationAdherence
- **Disease Focus:** Diabetes, Hypertension, COPD, Chronic Kidney Disease
- **Key Metrics:** HbA1c trends, BP variability, exacerbation frequency
- **Advanced Features:** Composite disease severity index, automatic follow-up scheduling
- **SQL Requirements:** Window functions for trend analysis, recursive CTEs for treatment pathways

Module 3: Pediatric Patient Clinical Data System

Primary Objectives:

- Design age-appropriate clinical database for pediatric patients
- Implement growth chart tracking with percentile calculations
- Create immunization schedule management with reminder system
- Develop developmental milestone assessment and tracking

Technical Specifications:

- **Entities:** PediatricPatient, GrowthMeasurement, Immunization, Milestone
- **Key Features:** WHO growth standards integration, catch-up immunization schedules
- **Age Groups:** Neonate (0-28d), Infant (1-12m), Toddler (1-3y), Child (4-12y)
- **Advanced Features:** Age-based medication dosage calculations, developmental delay alerts
- **SQL Requirements:** Age calculations in months/days, percentile ranking queries

Module 4: Geriatric Patient Health Record Database

Primary Objectives:

- Design comprehensive geriatric assessment database
- Implement fall risk assessment and prevention tracking
- Create cognitive function evaluation system
- Develop polypharmacy monitoring with interaction alerts

Technical Specifications:

- **Entities:** GeriatricPatient, FallRisk, CognitiveAssessment, Comorbidity

- **Assessment Tools:** MMSE, GDS, ADL/IADL scales, Morse Fall Scale
- **Key Features:** Frailty index calculation, caregiver burden assessment
- **Advanced Features:** Multi-morbidity complexity scoring, end-of-life care planning
- **SQL Requirements:** Complex scoring algorithms, medication count analysis

Module 5: Patient Allergy & Immunization Tracking System

Primary Objectives:

- Design comprehensive allergy profile management system
- Implement immunization schedule with due date calculations
- Create cross-reactivity database for drug allergies
- Develop adverse reaction documentation and reporting

Technical Specifications:

- **Entities:** Allergy, Immunization, Vaccine, AdverseReaction
- **Allergy Types:** Drug, Food, Environmental, Latex
- **Severity Levels:** Mild, Moderate, Severe, Anaphylaxis
- **Advanced Features:** Automatic contraindication checking, herd immunity calculations
- **SQL Requirements:** Date arithmetic for due dates, hierarchical allergy classification

Module 6: Clinical Alert System for Abnormal Vital Values

Primary Objectives:

- Design real-time vital sign monitoring database
- Implement Early Warning Score (EWS) calculation
- Create threshold-based alerting with escalation rules
- Develop trend analysis for vital sign deterioration

Technical Specifications:

- **Entities:** VitalSign, AlertRule, EscalationPathway, MonitoringDevice
- **Vital Signs:** Temperature, Heart Rate, BP, Respiratory Rate, SpO2, Pain Score
- **Scoring Systems:** MEWS, NEWS, PEWS (pediatric)
- **Advanced Features:** Multi-parameter trend analysis, predictive deterioration alerts
- **SQL Requirements:** Real-time aggregation, moving averages, pattern matching

Module 7: Symptom–Disease Mapping Database

Primary Objectives:

- Design comprehensive symptom-disease relationship database
- Implement weighted association strength calculations
- Create Bayesian diagnostic probability engine
- Develop differential diagnosis generation system

Technical Specifications:

- **Entities:** Symptom, Disease, SymptomDiseaseAssociation, DiagnosisRule
- **Data Sources:** ICD-11, SNOMED-CT symptom terminology
- **Association Metrics:** Sensitivity, Specificity, Likelihood Ratios
- **Advanced Features:** Symptom pattern recognition, diagnostic pathway optimization
- **SQL Requirements:** Probability calculations, set operations for symptom combinations

Module 8: Fever-Based Differential Diagnosis System

Primary Objectives:

- Design specialized fever evaluation database
- Implement fever pattern classification algorithms
- Create associated symptom matching system
- Develop urgency triage based on fever characteristics

Technical Specifications:

- **Entities:** FeverEpisode, TemperaturePattern, AssociatedSymptom, DifferentialDiagnosis
- **Patterns:** Continuous, Intermittent, Remittent, Hectic, Relapsing
- **Duration Categories:** Acute (¡7d), Subacute (7-21d), Chronic (¿21d)
- **Advanced Features:** Fever of Unknown Origin (FUO) workup guidance
- **SQL Requirements:** Time-series pattern analysis, temperature curve characterization

Module 9: Respiratory Symptom Diagnosis Database

Primary Objectives:

- Design respiratory symptom classification system
- Implement breath sound characteristic database
- Create smoking history and environmental factor integration
- Develop respiratory disease probability scoring

Technical Specifications:

- **Entities:** RespiratorySymptom, BreathSound, SmokingHistory, EnvironmentalExposure
- **Symptoms:** Cough, Dyspnea, Wheeze, Hemoptysis, Chest Pain
- **Cough Types:** Productive, Dry, Nocturnal, Post-tussive
- **Advanced Features:** Asthma control assessment, COPD exacerbation prediction
- **SQL Requirements:** Temporal symptom pattern matching, severity progression tracking

Module 10: Gastrointestinal Disorder Diagnosis Support

Primary Objectives:

- Design GI symptom assessment and correlation database
- Implement dietary habit tracking and symptom relationship
- Create stool characteristic analysis system
- Develop alarm symptom identification for serious conditions

Technical Specifications:

- **Entities:** GISymptom, DietaryRecord, StoolCharacteristic, AlarmFeature
- **Symptoms:** Abdominal pain, Diarrhea, Constipation, Nausea, Vomiting
- **Alarm Features:** Weight loss, Bleeding, Nocturnal symptoms, Family history
- **Advanced Features:** Rome Criteria for functional disorders, dietary trigger identification
- **SQL Requirements:** Symptom-diet correlation analysis, temporal pattern recognition

Module 11: Neurological Symptom Analysis Database

Primary Objectives:

- Design neurological symptom classification system
- Implement Glasgow Coma Scale tracking with trend analysis
- Create reflex and coordination test result database
- Develop neurological localization algorithms

Technical Specifications:

- **Entities:** NeurologicalSymptom, GCSScore, ReflexTest, Localization
- **Symptoms:** Headache, Seizure, Weakness, Numbness, Vision changes
- **Exam Components:** Cranial nerves, Motor, Sensory, Coordination, Gait
- **Advanced Features:** Stroke scale calculations, seizure frequency analysis
- **SQL Requirements:** Complex scoring algorithms, temporal event correlation

Module 12: Rule-Based Disease Ranking System

Primary Objectives:

- Design rule-based diagnostic inference engine
- Implement Bayesian probability calculations using SQL
- Create symptom severity weighting system
- Generate prioritized differential diagnosis lists

Technical Specifications:

- **Entities:** DiagnosticRule, ProbabilityTable, SymptomWeight, DiagnosisRank
- **Rule Types:** IF-THEN rules, Probabilistic rules, Temporal rules
- **Inference Methods:** Forward chaining, Bayesian updating
- **Advanced Features:** Diagnostic confidence scoring, alternative diagnosis suggestions
- **SQL Requirements:** Recursive rule evaluation, probability calculation functions

Module 13: Natural Language Patient Search System

Primary Objectives:

- Design keyword-based patient search engine
- Implement symptom-based patient cohort identification
- Create demographic filter combinations
- Generate SQL queries from natural language inputs

Technical Specifications:

- **Entities:** SearchQuery, QueryTemplate, SearchResult, PatientCohort
- **Search Types:** Demographic, Clinical, Temporal, Laboratory
- **Query Expansion:** Synonym matching, Concept hierarchy navigation
- **Advanced Features:** Relevance ranking, Search history learning
- **SQL Requirements:** Dynamic SQL generation, Full-text search optimization

Module 14: Clinical Query Translator for Laboratory Records

Primary Objectives:

- Design laboratory result query interpretation system
- Implement trend analysis queries for lab values
- Create abnormal result pattern identification
- Generate patient lists based on lab criteria

Technical Specifications:

- **Entities:** LabQuery, TrendPattern, AbnormalPattern, PatientCohort
- **Query Types:** Value-based, Trend-based, Pattern-based, Combination
- **Trend Analysis:** Slope calculation, Rate of change, Moving averages
- **Advanced Features:** Critical value alerts, Test sequence optimization
- **SQL Requirements:** Window functions for trends, Pattern matching queries

Module 15: Voice-Assisted Clinical Query System (Text Simulation)

Primary Objectives:

- Design text-based voice command simulation system
- Implement predefined query templates for common clinical questions
- Create contextual query understanding system
- Generate patient summaries from voice-style inputs

Technical Specifications:

- **Entities:** VoiceCommand, CommandTemplate, Context, SummaryTemplate
- **Command Types:** Retrieval, Calculation, Update, Summary
- **Context Awareness:** Patient context, Department context, Time context
- **Advanced Features:** Command history, Personalization, Error correction
- **SQL Requirements:** Template-based query generation, Context-aware filtering

Module 16: Doctor-Friendly SQL Query Dashboard

Primary Objectives:

- Design visual query builder interface simulation
- Implement common clinical question templates
- Design one-click report generation system
- Generate graphical data representation queries

Technical Specifications:

- **Entities:** Dashboard, Widget, ReportTemplate, Visualization
- **Widget Types:** Patient lists, Statistics, Trends, Alerts
- **Report Templates:** Daily rounds, Discharge summary, Quality metrics
- **Advanced Features:** Drag-and-drop query building, Real-time updates
- **SQL Requirements:** Aggregation queries, Time-series data extraction

Module 17: Smart Clinical Views using SQL

Primary Objectives:

- Design comprehensive clinical summary views
- Implement patient overview dashboards using views
- Create department-specific data presentations
- Generate real-time census and occupancy views

Technical Specifications:

- **Entities:** ClinicalView, Dashboard, DepartmentView, RealTimeData
- **View Types:** Patient summary, Department summary, Quality metrics
- **Refresh Strategies:** Real-time, Scheduled, On-demand
- **Advanced Features:** Materialized views for performance, View security
- **SQL Requirements:** Complex view definitions, View optimization techniques

Module 18: Question–Answering System for Hospital Database

Primary Objectives:

- Create FAQ-based query response system
- Implement common clinical scenario queries
- Design statistical question answering system
- Generate evidence-based answer retrieval

Technical Specifications:

- **Entities:** FAQ, QuestionTemplate, AnswerTemplate, Evidence
- **Question Types:** Factual, Statistical, Comparative, Temporal
- **Answer Formats:** Text, Table, Chart, Summary
- **Advanced Features:** Confidence scoring, Evidence linking
- **SQL Requirements:** Natural language processing simulation, Statistical calculations

Module 19: Drug–Drug Interaction Alert Database

Primary Objectives:

- Design comprehensive drug interaction database
- Implement severity-based interaction alerts
- Create mechanism of action conflict detection
- Generate alternative medication suggestions

Technical Specifications:

- **Entities:** Drug, Interaction, Alert, Alternative
- **Interaction Types:** Pharmacokinetic, Pharmacodynamic, Combined
- **Severity Levels:** Contraindicated, Major, Moderate, Minor
- **Advanced Features:** Risk-benefit analysis, Monitoring recommendations
- **SQL Requirements:** Matrix operations, Graph traversal for interaction chains

Module 20: Prescription Validation & Consistency System

Primary Objectives:

- Create prescription error checking system
- Implement dosage calculation validation
- Design frequency and duration consistency checks
- Generate prescribing guideline compliance reports

Technical Specifications:

- **Entities:** Prescription, ValidationRule, Guideline, Compliance
- **Validation Types:** Dosage, Frequency, Duration, Route
- **Guideline Sources:** National, Institutional, Specialty-specific
- **Advanced Features:** Cost-effectiveness analysis, Therapeutic substitution
- **SQL Requirements:** Complex constraint checking, Guideline compliance scoring

Module 21: Allergy-Aware Medication Alert System

Primary Objectives:

- Design patient allergy-medication cross-reference system
- Implement class-level allergy detection
- Create cross-reactivity alert mechanisms
- Generate safe alternative medication lists

Technical Specifications:

- **Entities:** Allergy, Medication, CrossReactivity, Alternative
- **Detection Levels:** Exact match, Class match, Cross-reactivity
- **Alert Types:** Contraindication, Caution, Information
- **Advanced Features:** Desensitization protocols, Graded challenge tracking
- **SQL Requirements:** Hierarchical drug classification, Similarity matching

Module 22: Polypharmacy Risk Detection Database

Primary Objectives:

- Create multiple medication risk assessment system
- Implement duplicate therapy detection
- Design therapeutic category overlap analysis
- Generate medication simplification recommendations

Technical Specifications:

- **Entities:** MedicationRegimen, RiskAssessment, Duplicate, Overlap
- **Risk Factors:** Number of medications, High-risk combinations, Age
- **Assessment Tools:** Beers Criteria, STOPP/START criteria
- **Advanced Features:** Deprescribing algorithms, Pill burden calculation
- **SQL Requirements:** Set operations for medication lists, Risk scoring algorithms

Module 23: High-Risk Drug Monitoring System

Primary Objectives:

- Design special monitoring requirement database
- Implement lab value monitoring for specific drugs
- Create dose adjustment rules based on patient parameters
- Generate monitoring schedule alerts

Technical Specifications:

- **Entities:** HighRiskDrug, MonitoringRequirement, LabAlert, DoseAdjustment
- **Drug Categories:** Anticoagulants, Chemotherapy, Immunosuppressants
- **Monitoring Types:** Laboratory, Clinical, Therapeutic drug monitoring
- **Advanced Features:** Personalized monitoring schedules, Escalation pathways
- **SQL Requirements:** Temporal monitoring schedules, Alert condition evaluation

Module 24: Automated Prescription Audit System

Primary Objectives:

- Create prescription quality assessment database
- Implement completeness and legibility checks
- Design guideline adherence scoring system
- Generate audit reports and quality metrics

Technical Specifications:

- **Entities:** PrescriptionAudit, QualityMetric, Violation, Improvement
- **Audit Areas:** Completeness, Legibility, Appropriateness, Safety
- **Quality Metrics:** Error rates, Compliance scores, Improvement trends
- **Advanced Features:** Root cause analysis, Targeted education suggestions
- **SQL Requirements:** Statistical quality analysis, Trend detection algorithms

Module 25: ICU Vital Signs Monitoring Database

Primary Objectives:

- Design high-frequency vital sign recording system
- Implement Early Warning Score (EWS) calculation
- Create trend analysis and prediction queries
- Generate deterioration alert algorithms

Technical Specifications:

- **Entities:** VitalSign, EWSScore, Trend, DeteriorationAlert
- **Monitoring Frequency:** Continuous, Hourly, 4-hourly as per acuity
- **Scoring Systems:** SOFA, APACHE, SAPS for ICU
- **Advanced Features:** Multi-organ dysfunction tracking, Response to intervention
- **SQL Requirements:** High-frequency data aggregation, Real-time scoring

Module 26: Emergency Room Patient Alert System

Primary Objectives:

- Create ER triage priority database
- Implement time-sensitive condition alerts
- Design resource allocation optimization queries
- Generate wait time and throughput reports

Technical Specifications:

- **Entities:** ERVisit, Triage, Alert, Resource, WaitTime
- **Triage Systems:** ESI, CTAS, MTS
- **Time Targets:** Door-to-doctor, Door-to-disposition, Length of stay
- **Advanced Features:** Crowding indices, Resource prediction, Flow optimization
- **SQL Requirements:** Queue management algorithms, Time interval calculations

Module 27: Cardiac ICU Monitoring Database

Primary Objectives:

- Design specialized cardiac parameter tracking
- Implement arrhythmia detection rules
- Create cardiac medication response monitoring
- Generate hemodynamic stability assessments

Technical Specifications:

- **Entities:** CardiacParameter, Arrhythmia, MedicationResponse, Hemodynamics
- **Parameters:** ECG rhythms, Cardiac enzymes, Echocardiogram findings
- **Scoring Systems:** TIMI, GRACE, HEART scores
- **Advanced Features:** STEMI recognition, Heart failure severity indices
- **SQL Requirements:** Pattern recognition for arrhythmias, Time-series analysis

Module 28: Neonatal ICU Monitoring System

Primary Objectives:

- Create specialized NICU parameter database
- Implement gestational age-adjusted normal ranges
- Design feeding and growth progression tracking
- Generate developmental care documentation

Technical Specifications:

- **Entities:** Neonate, GestationalAge, GrowthParameter, Feeding
- **Corrected Age:** Chronological vs corrected age calculations
- **Parameters:** Weight, Head circumference, Length, Vital signs
- **Advanced Features:** Intrauterine growth charts, Neurodevelopmental tracking
- **SQL Requirements:** Age correction calculations, Growth velocity calculations

Module 29: Threshold-Based Clinical Alert Database

Primary Objectives:

- Design customizable threshold management system
- Implement age and condition-specific thresholds
- Create escalation pathway definitions
- Generate threshold compliance reports

Technical Specifications:

- **Entities:** Threshold, PatientGroup, EscalationPath, Compliance
- **Threshold Types:** Absolute, Relative, Trend-based, Combination
- **Customization:** Patient-specific, Department-specific, Protocol-specific
- **Advanced Features:** Adaptive thresholds, Learning from outcomes
- **SQL Requirements:** Dynamic threshold application, Compliance monitoring

Module 30: Time-Series Patient Health Data System

Primary Objectives:

- Create temporal data analysis database
- Implement pattern recognition across time
- Design seasonality and trend detection
- Generate prognosis prediction queries

Technical Specifications:

- **Entities:** TimeSeries, Pattern, Trend, Seasonality
- **Time Scales:** Minutes, Hours, Days, Weeks, Months
- **Analysis Methods:** Moving averages, Rate of change, Cyclical patterns
- **Advanced Features:** Anomaly detection, Predictive modeling
- **SQL Requirements:** Window functions, Time-series aggregations, Pattern matching

Module 31: Historical Case Comparison Database

Primary Objectives:

- Design case similarity matching system
- Implement outcome-based case retrieval
- Create treatment effectiveness comparison
- Generate similar patient identification algorithms

Technical Specifications:

- **Entities:** HistoricalCase, SimilarityMetric, Outcome, TreatmentComparison
- **Similarity Dimensions:** Demographic, Clinical, Laboratory, Treatment
- **Matching Algorithms:** Exact match, Fuzzy match, Pattern match
- **Advanced Features:** Outcome prediction, Treatment optimization
- **SQL Requirements:** Similarity calculations, Pattern matching, Outcome analysis

Module 32: Treatment Outcome Analysis System

Primary Objectives:

- Create treatment response tracking database
- Implement outcome measurement standardization
- Design comparative effectiveness analysis
- Generate evidence-based treatment recommendations

Technical Specifications:

- **Entities:** Treatment, Outcome, Response, Recommendation
- **Outcome Types:** Clinical, Functional, Quality of life, Economic
- **Analysis Methods:** Before-after, Comparative, Cost-effectiveness
- **Advanced Features:** Personalized response prediction, Treatment sequencing
- **SQL Requirements:** Statistical analysis, Comparative queries, Trend analysis

Module 33: Disease Progression Case Repository

Primary Objectives:

- Design longitudinal disease tracking system
- Implement stage transition documentation
- Create complication prediction models
- Generate natural history databases

Technical Specifications:

- **Entities:** DiseaseProgression, Stage, Complication, NaturalHistory
- **Progression Models:** Linear, Non-linear, Stepwise, Fluctuating
- **Staging Systems:** Disease-specific staging (TNM, Child-Pugh, NYHA)
- **Advanced Features:** Progression rate calculation, Milestone prediction
- **SQL Requirements:** Temporal state transitions, Rate calculations, Pattern analysis

Module 34: Readmission Risk Analysis Database

Primary Objectives:

- Create readmission prediction scoring system
- Implement risk factor identification algorithms
- Design prevention strategy recommendations
- Generate quality improvement reports

Technical Specifications:

- **Entities:** Readmission, RiskFactor, PreventionStrategy, QualityMetric
- **Risk Models:** LACE, HOSPITAL, CMS models
- **Time Frames:** 7-day, 30-day, 90-day readmissions
- **Advanced Features:** Preventability assessment, Intervention effectiveness
- **SQL Requirements:** Risk score calculations, Temporal analysis, Intervention tracking

Module 35: Therapy Effectiveness Evaluation System

Primary Objectives:

- Design therapy response measurement database
- Implement side effect and benefit tracking
- Create cost-effectiveness analysis queries
- Generate personalized therapy recommendations

Technical Specifications:

- **Entities:** Therapy, Response, SideEffect, CostAnalysis
- **Effectiveness Measures:** Clinical improvement, Symptom relief, Survival
- **Safety Metrics:** Adverse events, Toxicity grades, Tolerability
- **Advanced Features:** Benefit-risk analysis, Quality-adjusted life years
- **SQL Requirements:** Benefit calculations, Cost analysis, Comparative effectiveness

Module 36: Similar Patient Case Retrieval System

Primary Objectives:

- Design patient similarity algorithm database
- Implement multi-parameter matching system
- Create treatment pathway comparison
- Generate prognostic insight delivery

Technical Specifications:

- **Entities:** PatientProfile, SimilarityAlgorithm, Match, Prognosis
- **Matching Parameters:** Demographic, Clinical, Genetic, Social
- **Similarity Measures:** Euclidean distance, Cosine similarity, Custom weights
- **Advanced Features:** Dynamic weighting, Context-aware matching
- **SQL Requirements:** Distance calculations, Multi-dimensional matching

Module 37: Secure Electronic Health Record (EHR) Database

Primary Objectives:

- Design comprehensive EHR with access controls
- Implement data encryption and security features
- Create audit trail for all data accesses
- Generate compliance with healthcare regulations

Technical Specifications:

- **Entities:** EHR, AccessControl, AuditTrail, SecurityPolicy
- **Security Features:** Encryption, Access logging, Data masking
- **Compliance:** HIPAA, GDPR, Local regulations
- **Advanced Features:** Break-glass access, Consent management
- **SQL Requirements:** Security views, Audit triggers, Encryption functions

Module 38: Role-Based Access Control for Hospital Systems

Primary Objectives:

- Design hierarchical access control system
- Implement role-permission mapping database
- Create emergency access override protocols
- Generate access violation reports

Technical Specifications:

- **Entities:** Role, Permission, User, AccessLog
- **Role Hierarchy:** Doctor, Nurse, Technician, Administrator, Patient
- **Permission Types:** Read, Write, Update, Delete, Export
- **Advanced Features:** Time-limited access, Context-based permissions
- **SQL Requirements:** Permission checking, Role inheritance, Access logging

Module 39: Clinical Audit Trail & Logging System

Primary Objectives:

- Design comprehensive activity logging database
- Implement tamper-evident audit records
- Create suspicious activity detection
- Generate compliance audit reports

Technical Specifications:

- **Entities:** AuditLog, Activity, User, SuspiciousPattern
- **Log Details:** Who, What, When, Where, Why
- **Security Features:** Digital signatures, Chain of custody
- **Advanced Features:** Anomaly detection, Predictive auditing
- **SQL Requirements:** Comprehensive logging triggers, Pattern detection

Module 40: Patient Consent & Data Privacy Database

Primary Objectives:

- Design consent management system
- Implement data sharing permission tracking
- Create research participation consent database
- Generate privacy compliance documentation

Technical Specifications:

- **Entities:** Consent, Permission, ResearchStudy, PrivacyPolicy
- **Consent Types:** Treatment, Research, Sharing, Marketing
- **Permission Levels:** Full, Limited, Anonymous, None
- **Advanced Features:** Dynamic consent, Granular permissions
- **SQL Requirements:** Consent tracking, Permission checking, Compliance reporting

Module 41: Multi-Role Access Control System

Primary Objectives:

- Design complex permission matrix database
- Implement temporary access granting system
- Create delegation of authority tracking
- Generate access review and recertification

Technical Specifications:

- **Entities:** User, Role, PermissionMatrix, Delegation
- **Matrix Features:** Role combinations, Context permissions, Time limits
- **Delegation Rules:** Hierarchical, Peer-to-peer, Emergency
- **Advanced Features:** Conflict resolution, Override tracking
- **SQL Requirements:** Complex permission checking, Delegation chains

Module 42: Secure Clinical Summary View Generator

Primary Objectives:

- Design context-aware summary generation
- Implement purpose-based data filtering
- Create anonymized data views for research
- Generate referral-specific summaries

Technical Specifications:

- **Entities:** Summary, Context, Purpose, Anonymization
- **Context Types:** Clinical, Research, Administrative, Legal
- **Anonymization Methods:** Pseudonymization, Aggregation, Perturbation
- **Advanced Features:** Dynamic summarization, Context adaptation
- **SQL Requirements:** View generation, Data transformation, Anonymization

Module 43: Laboratory Test Management Database

Primary Objectives:

- Design complete lab test catalog and ordering system
- Implement test preparation requirement documentation
- Create turnaround time tracking
- Generate test utilization reports

Technical Specifications:

- **Entities:** LabTest, Order, Preparation, Turnaround
- **Test Categories:** Hematology, Chemistry, Microbiology, Pathology
- **Preparation:** Fasting, Special handling, Timing requirements
- **Advanced Features:** Test combination optimization, Cost analysis
- **SQL Requirements:** Order tracking, Turnaround analysis, Utilization statistics

Module 44: Automated Lab Result Interpretation System

Primary Objectives:

- Design rule-based result interpretation engine
- Implement pattern recognition for result combinations
- Create diagnostic suggestion algorithms
- Generate follow-up test recommendations

Technical Specifications:

- **Entities:** LabResult, InterpretationRule, Pattern, Recommendation
- **Interpretation Types:** Normal, Abnormal, Critical, Pattern-based
- **Rule Types:** Single test, Multiple tests, Temporal patterns
- **Advanced Features:** Probability calculations, Confidence scoring
- **SQL Requirements:** Rule evaluation, Pattern matching, Recommendation generation

Module 45: Reference Range Validation Database

Primary Objectives:

- Design demographic-specific reference ranges
- Implement method and instrument-specific values
- Create dynamic range adjustment based on conditions
- Generate quality control violation alerts

Technical Specifications:

- **Entities:** ReferenceRange, Demographic, Method, Condition
- **Demographic Factors:** Age, Sex, Ethnicity, Pregnancy status
- **Method Variations:** Instrument, Reagent, Technique
- **Advanced Features:** Personalized ranges, Disease-specific adjustments
- **SQL Requirements:** Range selection algorithms, Dynamic adjustment

Module 46: Follow-Up Test Recommendation System

Primary Objectives:

- Design test sequence and dependency database
- Implement abnormal result follow-up protocols
- Create cost-effective test pathway suggestions
- Generate monitoring schedule recommendations

Technical Specifications:

- **Entities:** TestSequence, FollowUpProtocol, Pathway, Schedule
- **Sequencing Rules:** Prerequisite tests, Confirmatory tests, Monitoring tests
- **Protocol Types:** Disease-specific, Finding-specific, Risk-based
- **Advanced Features:** Cost-benefit analysis, Personalized schedules
- **SQL Requirements:** Path finding algorithms, Scheduling optimization

Module 47: Pathology Report Management Database

Primary Objectives:

- Design structured pathology reporting system
- Implement cancer staging and grading databases
- Create multidisciplinary team meeting documentation
- Generate treatment implication summaries

Technical Specifications:

- **Entities:** PathologyReport, CancerStage, Grade, MDTMeeting
- **Staging Systems:** TNM, FIGO, Ann Arbor, Dukes
- **Grading Systems:** Gleason, WHO, Bloom-Richardson
- **Advanced Features:** Prognostic scoring, Treatment response prediction
- **SQL Requirements:** Structured reporting, Staging calculations, Prognostic indices

Module 48: Critical Lab Value Alert System

Primary Objectives:

- Design panic value definition and management
- Implement rapid notification pathway system
- Create acknowledgment and action tracking
- Generate response time analysis reports

Technical Specifications:

- **Entities:** CriticalValue, NotificationPath, Acknowledgment, Response
- **Critical Levels:** Panic values, Alert values, Information values
- **Notification Paths:** Direct call, Text message, EHR alert, Escalation
- **Advanced Features:** Escalation protocols, Response verification
- **SQL Requirements:** Alert triggering, Notification tracking, Response timing

Module 49: Integrated Clinical Decision Support Database

Primary Objectives:

- Design comprehensive integrated clinical database
- Implement patient journey tracking across departments
- Create cross-domain decision support rules
- Generate holistic patient management views

Technical Specifications:

- **Entities:** IntegratedRecord, PatientJourney, CrossDomainRule, ManagementView
- **Integration Areas:** Clinical, Laboratory, Pharmacy, Radiology
- **Journey Tracking:** Timeline visualization, Department transitions
- **Advanced Features:** Unified patient view, Cross-system alerts
- **SQL Requirements:** Complex joins across domains, Integrated reporting

Module 50: AI-Inspired Medical Copilot using DBMS Concepts

Primary Objectives:

- Design complete medical assistant simulation
- Implement natural language query processing simulation
- Create evidence-based recommendation engine
- Generate clinical workflow optimization suggestions

Technical Specifications:

- **Entities:** CopilotSession, Query, Recommendation, Workflow
- **Assistant Functions:** Information retrieval, Decision support, Documentation
- **Learning Features:** User preference learning, Context adaptation
- **Advanced Features:** Predictive typing, Workflow integration
- **SQL Requirements:** Complex decision algorithms, Workflow optimization